

RELTWIN: CONTRASTING LOCALLY VALID TIMESTAMP ANSWERS FOR QUERY-SPECIFIC AUDIO GROUNDING

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ABSTRACT

Audio temporal grounding must distinguish which event occurrence answers a query, even when opposite event orders coexist in one recording. Models can instead return the same window for inverse queries, revealing a query–window binding error. We introduce RelTwin, which constructs opposite-order windows from shared audio excerpts and contrasts their timestamp answers under each query. Both windows have real acoustic support; the requested order determines the correct answer. This explicit answer-level competition complements sequence supervision and preserves direct timestamp generation at inference. Compared with paired SFT, RelTwin improves joint localization by 31.25 points on the synthetic relation-development set and retains an advantage under timing intervention. Across three training seeds, it averages 59.16 mIoU on SpotSound-Bench, surpassing the prior main-table best by 1.26 points.

Index Terms— Audio temporal grounding, timestamp answer contrasts, query–window binding, paired supervision

1. INTRODUCTION

Audio temporal grounding maps a query to event intervals [1, 2]. In a recording containing both “dog then rooster” and “rooster then dog,” the descriptions refer to different windows. Recognizing the categories establishes presence; answering the query requires selecting the requested order. Returning the same real window for both queries is a *query–window binding error*. We target this distinction between event detection and query-specific localization.

RelTwin (Fig. 1) addresses this binding problem through *locally valid timestamp contrasts*. Shared excerpts form opposite-order windows in one recording. Each window contains a real event sequence, but its correctness as an answer depends on the query. Sequence supervision rewards correct timestamp answers individually; the added candidate loss explicitly compares the two answers under each query. This local competition trains query–window binding while inference retains direct timestamp generation.

We make three contributions. First, we characterize query–window binding errors when inverse relations co-occur in one recording. Second, we introduce locally valid timestamp contrasts to supervise the query-specific choice between competing answers. Third, paired diagnosis and a timing intervention demonstrate improved query–window binding, alongside gains on SpotSound-Bench.

2. RELATED WORK

Audio temporal grounding. Weakly supervised grounding learns temporal correspondences from clip-level captions, reducing dependence on frame-level annotations [3]. Audio-language models such as Pengi [4], Qwen2-Audio [5], and Audio Flamingo 3 [6] provide audio-conditioned text generation. For temporal localization, TimeAudio introduces temporal markers and absolute time-aware encoding [7], while SpotSound interleaves timestamps with audio tokens and uses absent-event queries to suppress hallucinated intervals [8]. SpotSound thus combines event-presence supervision with timestamp prediction. RelTwin targets query-specific selection when inverse relations coexist: both descriptions have acoustic support in the recording, and the requested relation determines which occurrence is the answer.

Compositional audio–language learning. CLAP aligns audio and text in a shared representation space [9]. T-CLAP introduces order-reversed descriptions and a temporal-focused contrastive objective [10]. CompA evaluates event order and attribute binding, and trains CompA-CLAP with composition-aware hard negatives and modular contrastive learning [11]. CoSTALA combines spatio-temporal contrast with local event–caption alignment and feature consistency [12]. These methods organize audio–text representations around compositional relations, including their local structure. RelTwin places the supervision on complete timestamp answer sequences: their query-conditioned likelihoods determine the preference between competing windows. This trains relational selection in the output space of a generative grounding model.

Hard negatives for temporal grounding. In compositional video grounding, SHINE combines generated hard-

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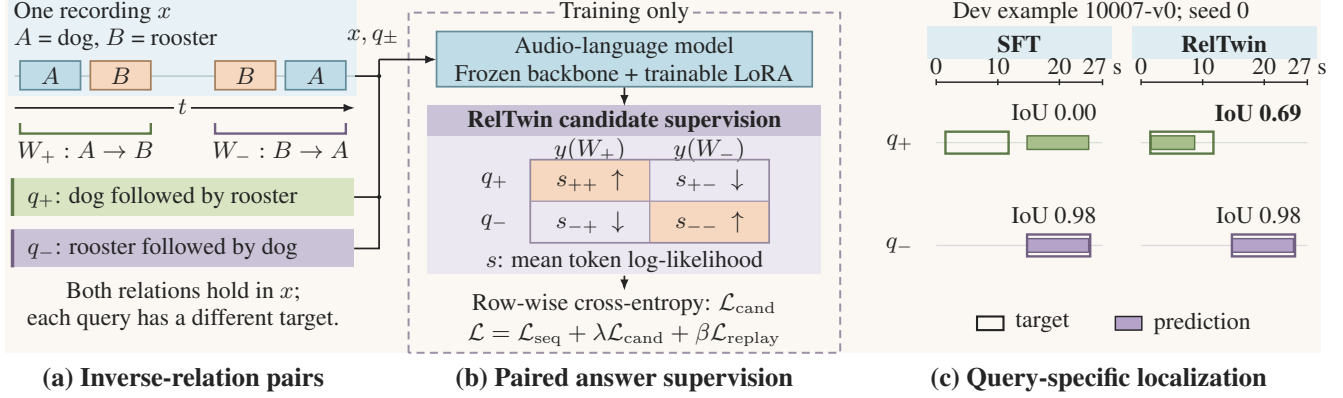


Fig. 1. RelTwin binds inverse queries to different windows in one recording. (a) Reused excerpts form two local targets. (b) The dashed enclosure marks training-only computation; shaded diagonal cells identify correct timestamp answers, and up/down arrows indicate training preferences. (c) Queries 56–57 (seed 0) illustrate SFT’s same-answer collapse and RelTwin’s query-specific localization. Outlines mark targets; filled bars show predictions, evaluated by set-IoU.

negative queries with within-video and cross-query saliency ranking [13]. Its training compares segment relevance as well as query compositions. RelTwin makes complete timestamp answers the candidates for comparison within one recording. Each candidate denotes an entire event sequence with real acoustic support. Under a fixed query, one answer matches the requested order and the other answers the inverse relation; reversing the query exchanges their labels while preserving the audio context. Comparing the length-normalized sequence log-likelihoods explicitly binds each relation query to its corresponding local answer.

3. QUERY-SPECIFIC WINDOW LEARNING

3.1. Two true relations, two different answers

RelTwin reuses two ESC-50 excerpts [14], A and B , to form windows $W_+ = (A \rightarrow B)$ and $W_- = (B \rightarrow A)$ in one recording x . Each spans the *entire ordered sequence*. For inverse queries (q_+, q_-) and candidates (W_+, W_-) , the answer-label matrix is $Y = I_2$ (Fig. 1). Both queries are present at recording level; the paired labels specify their local correspondence.

Each recording has two templates, “followed by” and “and then,” in both directions. Category queries target all occurrences of A or B for rehearsal. Alternating the earlier relation controls a constant-position preference; excerpt reuse fixes source material within a pair. Section 5.2 tests the contribution of layout-specific silence through a timing intervention.

3.2. Contrast timestamp answers during training

For the timestamp token sequence $y(S)$ describing interval set S , define its mean token log-likelihood

$$s_\theta(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_\theta(y_t | x, q, y_{<t}). \quad (1)$$

Candidates are $C = (W_+, W_-)$. Their relative answer preferences are $p_q(j) = \text{softmax}_j(s_\theta(x, q, C_j)/\tau)$. The paired candidate loss is

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_{q_+}(1) + \log p_{q_-}(2)]. \quad (2)$$

Reversing the query exchanges the positive timestamp sequence. Candidate cross-entropy makes this local discrimination explicit alongside correct-answer sequence supervision and category-query rehearsal:

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}. \quad (3)$$

Equation 3 defines *RelTwin*. SFT uses identical data and rehearsal with $\lambda = 0$. At inference, the model generates timestamps from audio and a query, without candidates or an additional scorer.

3.3. Joint localization and relation selection

For interval unions, $\text{IoU}_{\text{set}}(S, G) = |S \cap G|/|S \cup G|$. PairAcc requires both inverse-query predictions to achieve set-IoU ≥ 0.5 . We additionally require each to overlap its own target strictly more than the opposite target, yielding JointPairAcc. Swap error means that at least one query violates this strict preference, including ties.

Returning the union of two disjoint equal-length targets for *both* queries gives IoU 0.5 each: PairAcc passes but JointPairAcc fails. Requiring a preference for the query’s own

Table 1. Core training comparison (%). The seed-0 SFT/RelTwin pair shares training environment and updates. The backbone re-evaluation and three-seed summary provide separate references ($\pm$: sample SD). Joint: JointPairAcc.

Method	SpotSound mIoU	Relation development mIoU	Joint
Current SFT, s0	58.42	74.58	53.13
RelTwin, s0	59.43	88.55	84.38
RelTwin – SFT, s0	+1.02	+13.97	+31.25
SpotSound-A (re-eval.)	58.42	37.64	0.63
RelTwin, 3 seeds	59.16 $\pm$ 0.26	87.05 $\pm$ 1.43	81.46

Table 2. SpotSound-Bench results (400 queries, %). Published baselines: SpotSound Table 3 [8]. RelTwin reports three-seed means; Δ compares with the cited prior main-table best for each metric. Bold marks the highest listed point estimate.

Method	mIoU	R1@.3	R1@.5
WTATG	38.4	53.0	34.5
AM-DETR	19.5	25.5	15.0
Gemini-2.5-Flash	25.7	34.8	27.8
Gemini-2.5-Pro	19.7	20.0	17.3
Kimi-Audio	3.2	3.0	1.3
Qwen2-Audio	4.4	5.5	1.8
Audio Flamingo 3	7.6	8.8	2.8
TimeAudio	10.2	7.8	1.8
SpotSound-Q	49.9	65.0	51.3
SpotSound-A	57.9	76.5	59.5
RelTwin (3 seeds)	59.16	77.17	62.08
Δ vs. prior best	+1.26	+0.67	+2.58

target distinguishes relational selection from broad temporal coverage. Inspired by paired compositional evaluation [15], we report JointPairAcc alongside standard IoU to capture this behavior.

4. EXPERIMENTAL SETUP

Data and split roles. Adaptation uses 40 ESC-50 classes and 256 constructed recordings: 1,024 relation queries (512 pairs) and 512 rehearsal queries. The remaining 10 classes provide 80 recordings, 320 queries, and 160 inverse pairs. Adaptation/development classes and original Freesound IDs are disjoint; this separation is defined at the adaptation stage. Public evaluation covers all 400 SpotSound queries over 387 audios. The relation split and public scores have informed project development. We use the constructed split to diagnose query-window binding and SpotSound to measure performance on public grounding queries.

Adaptation and the primary control. Both meth-

ods retain SpotSound-A’s LoRA adaptation [16] on Audio Flamingo 3, with rank 8, scaling parameter $\alpha = 16$, dropout 0.1, and 20,185,088 trainable adapter parameters. SFT and RelTwin use 256 AdamW [17] updates, zero weight decay, learning rate 5×10^{-6} , gradient clipping 1.0, $\tau = 1$, and $\beta = 0.5$; RelTwin uses $\lambda = 1$. Each update processes one inverse-query pair (two relation queries) and one category-rehearsal query. Gradients accumulate over these components, with no accumulation across pairs. The primary SFT–RelTwin comparison uses seed 0 with the same software environment, code, initial weights, and data order. Updates are matched; FLOPs and wall time are not. RelTwin scores four query–answer combinations without gradients then recomputes weighted gradients, whereas SFT backpropagates two positive answers. Both add rehearsal.

We also report RelTwin seeds 0/1/2, using final checkpoints and including every seed. SpotSound-A is re-evaluated with the same inference code as the adapted models to provide a backbone reference. All evaluations use 16 kHz mono audio, bfloat16 inference, and greedy decoding (one beam, no sampling) with at most 128 new tokens per query. The three-seed RelTwin summary is separate from the matched seed-0 SFT comparison.

Uncertainty and reproducibility. We verify row alignment, audio identity, duration, and recomputed IoU. Bootstrap uses 20,000 audio-group draws, keeping queries together. Relation CIs additionally resample 39 original-source connected groups (largest: 18 audios) to address excerpt reuse. The intervals quantify sample uncertainty conditional on observed seeds and do not adjust for development selection. Code, inputs, weights, and environment are hashed before inference.

5. RESULTS AND ANALYSIS

5.1. Candidate supervision improves binding

RelTwin improves both relation selection and temporal overlap in the primary SFT comparison (Table 1). Relation mIoU increases from 74.58 to 88.55 (+13.97 points), with source-group 95% CI [9.84, 17.51]. JointPairAcc rises from 53.13 to 84.38 (+31.25 points; CI [22.35, 40.50]), and Swap error falls from 40.00 to 7.50. On SpotSound, SFT scores 58.42 and RelTwin 59.43: +1.02 mIoU points with audio-group CI [0.23, 1.81]. Shared positive examples and update counts connect these gains to adding explicit competition between timestamp answers. Across the 400 public queries, the paired win/tie/loss counts are 92/239/69, including two IoU decreases of at least 0.5.

Paired diagnosis reveals a change in answer selection. SFT returns identical answers for 37/160 inverse-query pairs (22 audios), compared with 1/160 for RelTwin; 27 of those 37 pairs become jointly correct. Overall, 51 pairs change from incorrect to correct, one in the opposite direction, and

24 remain incorrect. Figure 1 illustrates this recovery: SFT returns the same interval for both queries, whereas RelTwin selects the corresponding relation window for each query. The measured set-IOU values retain the distinction between relation selection and boundary accuracy. JointPairAcc ties the changed answers to the correct relation targets.

5.2. Binding gains persist after timing intervention

The timing intervention tests whether layout-specific silence explains the binding advantage. We recompose all 80 development audios with 1.5 s lead-in, 0.25 s within-window gap, 3 s between windows, and 2 s tail, preserving event waveforms, order, and queries. Frozen SFT/RelTwin seed 0 models evaluate one template (160 queries). Forty AB-first audios remain waveform-identical and reproduce every prediction; the 40 BA-first audios receive the changed timing layout.

On the changed audios, RelTwin achieves 87.50 JointPairAcc versus 67.50 for SFT: +20.00 points, source-group CI [2.94, 33.33]. Under the original layout, the corresponding scores are 85.00 and 42.50. Uniform timing raises SFT’s score and reduces the gap from 42.50 to 20.00 points, while the positive interval shows a retained binding advantage. Across all 160 queries, uniform-timing mIoU is 81.34 for SFT and 90.65 for RelTwin. This targeted, post-development test shows that the original silence pattern does not fully explain the answer-discrimination advantage.

5.3. Public grounding performance

RelTwin averages 59.16 SpotSound mIoU across three seeds, exceeding the cited prior main-table best of 57.9 by 1.26 points (Table 2). R1@.3 and R1@.5 rise by 0.67 and 2.58 points, linking answer discrimination to overlap and recall gains on public queries. Literature scores provide performance references; Table 1 supplies the SFT contrast. Against the current official re-evaluation at 58.42 mIoU, RelTwin gains 0.75 points, with audio-group CI [−0.24, 1.74] spanning zero.

5.4. Scope

Repeated excerpts and fixed templates enable controlled binding tests. Alternative hard-negative constructions, matched multi-seed replication, and natural overlapping or long-range relations remain open evaluations. Adaptation-stage source separation does not audit backbone pretraining overlap.

6. CONCLUSION

Event presence alone does not determine which interval matches a relation query. RelTwin trains query–window binding through locally valid timestamp contrasts: competing windows have real acoustic support, but their correctness

depends on the query. Paired diagnosis and the timing intervention show improved joint localization over paired SFT, alongside gains on SpotSound-Bench. These results support explicit query-conditioned competition between timestamp answers while retaining direct timestamp generation at inference.

7. ACKNOWLEDGMENTS

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8. COMPLIANCE WITH ETHICAL STANDARDS

This computational study reuses public audio from ESC-50 [14] and SpotSound-Bench [8], including programmatically constructed sequences. No new recordings, participant recruitment, human or animal experiments, or listening studies were undertaken.

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